



INTERRUPTS

Breaks That Make Your PC Run

Computers have become a vital part of our life today. It would be difficult to imagine a world without them. Using a PC, we perform a multitude of tasks quickly and accurately. Have you ever noticed how many times you press a key on the keyboard, or how often you click your mouse? Ever wondered how the PC is able to keep track of all these inputs and handle them? This is closely related with interrupts. We interrupt our PC numerous times in a day, without realising what goes on behind the scenes. This article explains the concept of interrupts.

*I*n simple English, an interrupt means to cause, or make a break in the continuity or uniformity of a course or a process. Similarly, in computer science, an interrupt is a signal from hardware or software, indicating the need for the CPU's attention; i.e., interrupts provide a way to get the CPU's attention.

Let's look at the example of your keyboard. It connects to the PC via a keyboard controller. Whenever you press a key, the keyboard sends the keyboard controller a scan code, which is then passed by the keyboard controller to the CPU (see Figure 1). The keyboard controller can hold only a single keystroke. Therefore, the keyboard controller must be freed before